The data: *toxo* infection status and whether in a car wreck or not.

		Observed	
	N	Not infected	Infected
Car wreck	185	124	61
Not in car wreck	185	169	16

1. Which variable is conditional (dependent) on the other?

Car wreck is conditional on toxo infection

2. Explain your answer to the question above.

Logic… a brain parasite could influence behavior / driving ability. We would not expect toxo infection to depend on getting in a car wreck.

3. A) Use the data above to calculate the probability of being infected with *Toxoplasma*. Round to the nearest hundredth (e.g. X.XX)

77/(185*2) = 0.21

   B) Use the data above to calculate the probability of being in a car wreck. Round to the nearest hundredth (e.g. X.XX)

   185/(185*2) = 0.5

4. Write out the equation that would allow you to calculate the expected number of individuals that are infected and in a car wreck.

Prob car wreck * prob infection * total number of people = 0.21 * 0.5 * 370

5. Calculate the expected number of individuals that are infected and in a car wreck. Round to the nearest hundredth (e.g. XX.XX).

   0.21*0.5*370 = 38.85

6. What is the chi-square value for infected car wreckers? Write out the math, round to the nearest hundredth (e.g. XX.XX).

$(obs - exp)^2 / exp = (61-38.85)^2 / 38.85 = 12.63$

7. What is the p-value associated with the chi square value?

$p < 0.001$

8. Use the results of your analysis to assess whether the probability of being in a car wreck is conditional on *toxo* infection. Within your claim, state the null hypothesis and whether or not it is refuted by the results.

   Example: The results show that getting in a car wreck is conditional on *toxo* infection. We can reject the null hypothesis that these two variables are independent from each other - the probability of the difference between the observed and expected number of infected car wreckers being due to sampling error is very small ($X^2 = 12.63$, $p < 0.001$).